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St. Joseph's College of Arts & Science (Autonomous)
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CH510 - INORGANIC CHEMISTRY-III

Time : 3 hrs **Max Marks :75**

SECTION – A (20X1=20)

Answer **ALL** Questions

I. Choose the correct answer

1. Among the following, which can be used to estimate Ni^{+2}
   a) DMG b) OH^- c) Cl^- d) S^{-2}
2. The coordination sphere of a complex consists of _____.
   a) The central metal ion only
   b) The ligands
   c) The central metal ion and the ligands bonded to it
   d) The primary and secondary valencies

3. Which of the following statements is (are) false?
   a) The greater the energy gap in a metal complex, the shorter the wavelength of light the complex will absorb.
   b) Complex color depends on both the metal and the ligand.
   c) Metal complexes with an ammonia ligand have a larger energy gap than the corresponding fluoride complexes.
   d) Strong field ligands are associated with low energy gaps.
   e) Both A and C

4. What is the charge on the complex ion in $Mg_2[FeCl_6]$?
   a) 2- b) 2+ c) 3- d) 3+ e) 4-

5. Which one of the following species is paramagnetic?
   a) Cu^+ b) Cr^{3+} c) Zn d) Ca e) Ag^+

6. The correct name for $[Ni(NH_3)_6](NO_3)_3$ is _____.
   a) Dinitrohexaamminenickel (II)
   b) Hexaamminenickel (III) trinitrate
   c) Dinitrohexaamminenickelate (III)
   d) Hexaamminenickel (II) nitrate
   e) Hexaamminenickel (III) nitrate

7. The complex $[Zn(NH_3)_2Cl_2]^{2+}$ does not exhibit cis-trans isomerism. The geometry of this complex must be _____.
   a) Tetrahedral b) Trigonal bipyramidal c) Octahedral
   d) Square planar e) Either tetrahedral or square planar

8. Linkage isomerism can only occur _____.
   a) In cis-isomers of octahedral complexes
   b) With cobalt complexes
   c) With coordination number 6
   d) With tetrahedral complexes
   e) With ligands that have more than one possible donor atom

9. Which of the following will display optical isomerism?
   a) Square-planar $[Rh(CO)_2Cl_2]^-$
   b) Square-planar $[Pt(H_2NC_2H_4NH_2)_2]^{2+}$
   c) Octahedral $[Co(NH_3)_6]^{3+}$
   d) Octahedral $[Co(NH_3)_5Cl]^{2+}$
   e) Octahedral $[Co(H_2NC_2H_4NH_2)_3]^{3+}$

10. Consider a complex in which manganese (II) is bonded to six identical ligands. Which one of the following ligands will result in the smallest value of Δ?
    a) Cl^- b) NH_3 c) H_2O d) F^- e) CN^-

II. Fill up the blanks:

11. The most common coordination numbers are _____.

12. Six-coordinate complexes generally have _____ geometry.

13. The chelate effect is enhanced by polydentate ligand binding because of the change in _____.

14. Name the compound: $K_2[Cr(H_2O)_4(CO_3)_2]$ is ----------

15. The CFSE of highspin d^5 system is ----------

III. Match the following:

16. tn	i. Geometrical isomerism
17. SCN-	ii. Optical isomerism
18. H2O	iii. Ligand isomerism
19. [Co(en)3] 3_+	iv. Linkage isomerism
20. Square planar complexes	v. Solvate isomerism

SECTION – B (10X2=20)

Answer any **TEN** Questions

21. A double salt is not a co ordination compound. Why?

22. Which of the two is more stable K4[Fe(CN)6] or K3[Fe(CN)6].Why?

23. Give two examples for bidentate ligands.

24. Write a note on polymerisation isomerism.

25. Calculate the EAN of Co in $[Co(NH_3)_6]^{3+}$
26. Give any two limitations of VBT.

27. Identify the spin states of the following complexes depending upon the type of hybridization.i)$[Co(NH_3)_6]^{3+}$ ii) $[Co(F)_6]^{3-}$

28. How will you estimate Aluminium using Oxine.

29. What is spectrochemical series?Why is it called so?

30. Draw the structure of rutile.

31. What are superconductors? Give one example.

32. Give any two consequences of Frenkel defect.

SECTION- C (5X7=35)

Answer any **FIVE** Questions

33. Explain the following with suitable examples.
    a) Ionization isomerism b) Linkage isomerism
    c) Ligand isomerism d) Co-ordination isomerism **(2+2+2+1)**

34. Account for the following:
    a) Octhedral field splitting energy,$\Delta_o$is always higher than tetrahedral field energy,Δ_t. **(3)**
    b) $[Mn(H_2O)_6$ is a high-spin complex whereas $[Mn(CN)_6]^{4-}$ is a low spin complex. **(2+2)**

35. Explain the hybridisation & geometry of $Cr(CO)_6$ &$Mn_2(CO)_{10}$**(3+4)**

36. a) How Schottky & Frenkel defects are created. **(4)**
    b) Derive Bragg's equation. **(3)**

37. a) Derive the relationship between stepwise&overall stability constants. **(3)**
    b) How will you determine the stability constant by spectrophotometric method. **(4)**

38. a) For an octahedral Mn 3+ complex,Δo =21150 $cm^{-1}$and electron pairing energy Ep=27750 cm^{-1}.Will it be a high spin or low spin complex?Calculate its CFSE.Find out the spin only magnetic moment of the complex. **(3+2)**
    b) Define trans effect. **(2)**

39. a) $[Co(CN)_6]^{3-}$ is diamagnetic but$[CoF_6]^{3-}$ is paramagnetic with a moment of about 5.3 BM.Account for the difference in magnetic moment of these two octahedral complexes using CFT. **(3)**
    b) Using CFT 'show the orbital occupancyies for both strong and weak octahedral field for Mn^{2+},Fe^{2+} **(2)**
    c) State the Jahn Teller effect **(2)**

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